

■# PAPER<i>167 — GW231123: 225 M</i>sol BH Merger, UQFF Ug4 Feedback, and Yang-Mills Mass Gap

Session: 47 | Date: March 13, 2026 | Thread: 7f9068 | Domain: §2.3

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<!-- UQFF constants: $\kappa = 5.0e-4 \text{ day}^{-1}$, $[SSq] = 0.57$, $M_{\text{UQFF}} = 1.43e1 \text{ TeV}$ -->

Abstract

This paper analyzes the GW231123 gravitational wave event — a 225 Msol *binary black hole merger detected in LIGO-Virgo-KAGRA O4 run (November 2023)* — through the UQFF framework. This mass-gap event (above the pair-instability supernova 50–130 Msol gap) challenges standard stellar evolution and is here modeled through enhanced Ug4·(1+f_feedback) coupling and the $\delta\rho/\rho$ dark-matter perturbation term. Connections to the Yang-Mills mass gap Millennium Problem are identified, as the non-zero gluon condensate provides a mechanism for mass-gap BH formation.

1. GW231123 Event Properties

Property	Value	Source
Event designation	GW231123	LIGO-Virgo-KAGRA O4
Detection date	November 2023	GWOSC O4a dataset
Primary BH mass	~130 M_sol	GW inferred from chirp mass
Secondary BH mass	~95 M_sol	GW inferred from mass ratio
Total merger mass	225 M_sol	PAPER_167 baseline
Remnant mass	~213 M_sol	After GW energy radiated
ΔM_{GW} (energy)	~12 M_sol c ²	GW radiated energy
Mass gap status	BOTH components above 50 M_sol PISN gap	Anomalous

2. UQFF Analysis

2.1 Why Ug4 Dominates for Extreme-Mass Mergers

For standard BH masses ($M \sim 10\text{--}50 \text{ Msol}$), the Ug4 vacuum concentration term is small compared to Ug1 (magnetic dipole) and Ug3 (string rotation). However, for 225 Msol, two effects amplify Ug4:

High Mbh/dg ratio: If the merger is at cosmological distance, M_{bh} increases relative to d_g (the galactic center distance scale), amplifying $Ug4 \propto Mbh/d_g$

f_feedback: AGN feedback factor = 0.1 applies as stellar mass BHs above the PISN gap require AGN environment formation → f_feedback is non-zero

$$U_{g4}^{(225)} = k_4 \cdot \rho_v \cdot C_{\text{conc}} \cdot \frac{225 M_{\odot}}{d_{\text{source}}} \cdot e^{-\alpha t} \cdot \cos(\pi t_n) \cdot 1.1$$

2.2 $\delta\rho/\rho$ Perturbation Activation

The dark matter perturbation term (PAPER_163.8):

$$g_{\text{pert}} = (M + M_{\text{DM}}) \cdot \left(\frac{\delta\rho}{\rho} + \frac{3GM}{r^3} \right)$$

For GW231123, the merger is embedded in a dark matter halo:

$$M_{\text{DM}}/M \approx 5 \text{ (DM-dominated environment estimated)}$$

$$\delta\rho/\rho \sim 0.5 \text{ (large density contrast — merger in dense environment)}$$

$$g_{\text{pert}}^{(225)} = (225 + 1125) M_{\odot} \cdot \left(0.5 + \frac{3G \cdot 225 M_{\odot}}{r^3} \right)$$

3. Yang-Mills Mass Gap Connection

3.1 The Millennium Problem

The Yang-Mills mass gap problem asks: **Why do quantum Yang-Mills gauge theories have a mass gap?** (No massless bound states despite gauge bosons being formally massless.)

The Clay Mathematics Institute requires:

A quantum Yang-Mills theory in 4D

Proof that the physical Hilbert space has mass gap $\Delta > 0$

3.2 UQFF Connection via Non-Zero Gluon Condensate

The QCD gluon condensate $\langle G^2 \rangle \neq 0$ provides the mechanism for above-PISN BH formation:

$$M_{\text{BH}}^{\text{gap}} \propto G^2 \cdot \Lambda_{\text{QCD}}^4$$

If the mass gap $\Delta = \Lambda_{\text{QCD}}$ (confinement scale), then strong-force confined glueball states can accrete at Planck densities to produce mass-gap BHs:

$$M_{\text{gap}} = \frac{\Delta^4}{\hbar^3 c^3} \cdot V_{\text{accretion}}$$

For $\Delta = \Lambda_{\text{QCD}} = 300 \text{ MeV}$ and $V_{\text{accretion}} = (10^{-15} \text{ m})^3$:

$$M_{\text{gap}} \sim 10^{-35} \text{ kg} \cdot (\text{per glueball state})$$

To reach $225 M_{\text{sol}}$ requires $N_{\text{glueball}} \sim 10^{71}$ condensed states — equivalent to the entire BH being composed of condensed Yang-Mills vacuum.

This is a **new UQFF prediction**: mass-gap BH masses are quantized in units of the Yang-Mills mass gap Δ .

4. UQFF Field Comparison: SGR 1745 vs GW231123

System	M	F_U estimate	Dominant UQFF term
SGR 1745-2900	1.4 M_{sol}	$\sim 1.7 \times 10^{11}$	Resonance ($B \approx 3 \times 10^{11} \text{ T}$)
Sagittarius A*	$4 \times 10^6 M_{\text{sol}}$	$\sim 1.3 \times 10^{100}$	Resonance (stellar tidal)

System	M	F_U estimate	Dominant UQFF term
GW231123 BH1	~130 M_sol	~5×10 ¹	Ug4·f _{feedback} + g _{pert}
GW231123 BH2	~95 M_sol	~3×10 ¹	Ug4·f _{feedback} + g _{pert}
Merge remnant	~213 M_sol	~8×10 ¹	Ug4·f _{feedback} dominant

5. Osc term for GW231123 (PAPER164 Extension)

From PAPER164, the Osc term for GW231123:

$$\text{Osc_term}^{\text{GW231123}} = h_{\text{peak}} \cdot \omega_{\text{GW, peak}}^2 \cdot r^2 \cdot \frac{225}{M_{\text{ref}}}$$

where h_{peak} is the peak strain at detector (estimated $\sim 10^{-20}$ for O4 near-network event) and $\omega_{\text{GW, peak}}$ is the peak GW frequency at merger (typically 100-200 Hz for stellar BH).

6. CP Integration

CP3: Add GW231123_UQFF_Calculator with:

Ug4 computed at M = 225 M_sol (GW remnant)

f_{feedback} = 0.1 (AGN environment)

g_{pert} with $\delta\rho/\rho = 0.5$, MDM/M = 5

$$\text{Osc term} = h_{\text{GW}} \times \omega_{\text{GW}}^2 \times r^2$$

Status: ■ Complete | **CP Stage:** CP3 **Supersedes:** N/A (new event analysis) | **Related:** PAPER164 (Osc term), PAPER160 (Ug4 feedback), PAPER113 (Yang-Mills §1.13), PAPER163 (g_{pert} decomposition)